

Danush Shekar

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Education

2022 – Present	Ph.D. Candidate, Physics University of Illinois, Chicago, <i>USA</i> Supervisor: Dr. Corrinne Mills
2017 – 2022	Integrated M.Sc., major in Physics, minor in Mathematics National Institute of Science Education and Research, <i>Bhubaneswar, India</i> Supervisor: Dr. Bedangadas Mohanty Thesis: Simulations and Prototyping of an MPGD and a Wire-based detector

Awards and Honors

2025	URA Fellowship Research fellowship to develop simulation datasets and ASIC test flows for smartpixels at Fermi National Accelerator Laboratory, USA
2025, 2026	UIC Graduate Student Council Travel Award Awarded funding to present research at the APS Global Physics Summit, 2025 and TIPP, 2026
2022	Best Master's Thesis Award Awarded to the best M.Sc. thesis in the graduating class of the Physics department at NISER, India
2017 – 2022	Scholar, DAE Incentive Scheme for Holistic Science Education and Augmentation Supported by the Department of Atomic Energy, India
2020	Tata Institute of Fundamental Research's Visiting Students' Research Program Selected for a funded internship program for research at TIFR, India
2019	Finalist, PRL's Vikram Sarabhai Innovation CompetitiON (VISION) One of top 11 students in the country to propose a scientific experiment for funding by Physical Research Laboratory, India
2016	Scholar, Next Genius Foundation (declined) One of top 35 students in the country to receive a scholarship offer to study at The Wheaton College, MA, USA

Selected publications

Planar silicon + edge/hardware AI (TCAD, ASIC-testing, ML)

D. Shekar, B. Weiss, et al. Sensor Co-design for smartpixels. NIM-A, under review (2026)

B. Parpillon, A. Badea, D. Shekar et al. In-pixel integration of signal processing and AI/ML based data filtering for particle tracking detectors. Under preparation (2026)

AC-LGAD technology (TCAD, laser & proton beam tests)

D. Shekar, S. Nanda, et al. Studying AC-LGAD strip sensors from laser and testbeam measurements. JINST (2026)

I. Dutta, C. Madrid, R. Heller, S. Nanda, D. Shekar, et al. Results for pixel and strip centimeter-scale AC-LGAD sensors with a 120 GeV proton beam. NIM-A (2025)

MAPS technology (Simulation, proton beam tests)

Characterisation of the first wafer-scale prototype for the ALICE ITS3 upgrade: The monolithic stitched sensor (MOSS). NIM-A (2026)

Conference proceedings

Smartpixels: radiation-hard ASICs with on-chip neural networks in 28 nm CMOS. Under preparation, NIM-A, TIPP (2026)

Characterization of a prototype single ThGEM detector. DAE Symp. on Nucl. Phys. 66 (2022) 1148-1149, 66th DAE Symposium on nuclear physics (2022)

Conference presentations

Smartpixels: radiation-hard ASICs with on-chip neural networks in 28 nm CMOS. TIPP, Mumbai, India (2026)

(Invited) AI on ASICs: SmartPixels. EPIGRAPHY, Lisbon, Portugal (2026)

Smartpixels: Developing ASICs for high-energy particle detectors with on-chip neural networks in 28 nm CMOS. CPAD, Philadelphia, USA (2025)

Sensor co-design for Smartpixels. CPAD, Knoxville, USA (2024)

Performance of AC-LGAD strip sensors with laser and 120 GeV proton beam. APS Global Physics Summit, Virtual (2024)

Presented a talk and poster (One of 130 students from 4 countries selected to discuss and develop measures towards environmental issues for a one-year implementation phase).

Greening with Goethe International Environmental Youth Conference, Bengaluru, India (2012)

Schools attended

2024	MAPS Academy (Among 20 selected students from around the world) KEK, Tsukuba, Japan
2024	CMS - Data Acquisition School Fermi National Accelerator Laboratory, Batavia, USA
2023	Excellence in Detector and Instrumentation Technologies Brookhaven National Laboratory, New York, USA

Teaching and Mentorship

2025 – Present	APS IMPact mentor Virtual Helping undergraduate students navigate through career options
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2025	Teaching assistant, General Physics I University of Illinois, Chicago, <i>USA</i> Taught and graded this course for undergraduate students
2024	Teaching assistant, Problem solving workshop University of Illinois, Chicago, <i>USA</i> Developed and taught problem-solving sessions for physics majors
2024	Teaching assistant, Vibrations, Waves, and Thermal Physics University of Illinois, Chicago, <i>USA</i> Taught and graded this laboratory course for physics majors
2023	Teaching assistant, Physics for Life Sciences University of Illinois, Chicago, <i>USA</i> Taught and graded the laboratory component of the course
2022 – 2023	Teaching assistant, Modern Experimental Physics (undergraduate and graduate level) University of Illinois, Chicago, <i>USA</i> Taught and helped develop the laboratory curriculum for the course, which includes experiments in modern physics and electronics

Professional service

2026 – Present	Reviewer, Nuclear Instruments and Methods in Physics Research Section A
2020 – 2022	Representative for Undergraduate Committee of the School (Physical Sciences) NISER, <i>Bhubaneswar</i>
2019 – 2020	Secretary of Hostel Executive Council NISER, <i>Bhubaneswar</i>
2018 – 2019	Coordinator of Weekly Talks, NISER Astronomy Club NISER, <i>Bhubaneswar</i>

Technical Skills

Finite element method and device simulation tools:

Silvaco TCAD, Synopsys Sentaurus, PixelAV, Garfield++, COMSOL Multiphysics

Programming:

Python, C++

Design:

AutoCAD, Blender, Adobe Illustrator, HTML5, CSS3

Last updated April 15, 2026